

Measuring the Transverse Beam Profile of a Helium-Neon Laser Using a Newtonian Telescope

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A Gaussian laser is a theoretical device that exhibits characteristics of an ideal laser beam, such as having a normal distribution of intensity^{1,2}. Using a Gaussian laser as a standard when analyzing the behavior of actual beams can assist in understanding various laser properties, such as their intensity profiles and beam radii. Knowing such factors can lead to gaining greater knowledge of how to operate and apply different calibers of laser in various situations. Our objective for this lab was to determine the beam width and transverse profile of a helium-neon laser beam using the razor blade method and a Newtonian Telescope. The primary outcome we aimed to derive from this research is the beam width before and after expansion, as well as the determination of how the experimental calculation of magnification compares with the predetermined magnification results.

To measure the transverse beam profile of an expanded helium-neon laser, we designed a Newtonian telescope with 200 mm and 50 mm biconvex lenses to magnify the beam by 4x. A 75 mm plano-convex lens was also added to our optical setup to focus the expanded beam to a narrow point in front of the analyzer. To conduct the experiment, a razor blade was placed on a mount and oriented perpendicular to the plane of incidence. At the beginning of the data collection process, the beam was originally unobstructed; the razor blade was then adjusted in small increments until the beam was entirely covered. To measure the transverse profile of the unexpanded beam, the Newtonian telescope was removed, and the beam was realigned. The original profile of the expanded and unexpanded beam was fitted to the complementary error function,

$$G(x) = I \cdot \operatorname{erfc}\left(\frac{\sqrt{2}(x-x_0)}{w}\right) + I_b \quad (1)$$

Where I is the intensity of the beam, I_b is the intensity of the background, w is the beam width, and x_0 is the initial position. In taking the derivative of both transverse beam profiles relative to their position, figure 2 (b) was obtained and fitted to the Gaussian function,

$$g(x) = I_0 e^{\frac{-2(x-x_0)^2}{w^2}} \quad (2)$$

After fitting each figure to its respective function, the beam width of the laser before and after expansion, along with the magnification, was determined and is listed in the table below:

	w_i (mm)	w_f (mm)	m	Relative % Error of m
Error Function Fit	0.912 ± 0.046	3.721 ± 0.103	4.082 ± 0.235	2.038
Gaussian Function Fit	0.964 ± 0.099	4.035 ± 0.404	4.186 ± 0.602	4.645
Interpolated Data	0.965 ± 0.063	3.992 ± 0.252	4.135 ± 0.376	3.378

Table 1: Beam widths of a helium-neon laser beam obtained from fitting the beam's transverse profile and its derivative before and after expansion, along with each experimentally determined factor of magnification

¹ Edmund Optics, "Gaussian Beam Propagation," [Edmundoptics.com](https://www.edmundoptics.com), (2017).

² Dr. Rüdiger Paschotta, "RP Photonics Encyclopedia - Gaussian beams, laser beam, fundamental transverse modes," [Rp-Photonics.com](https://www.rp-photonics.com), (2019).

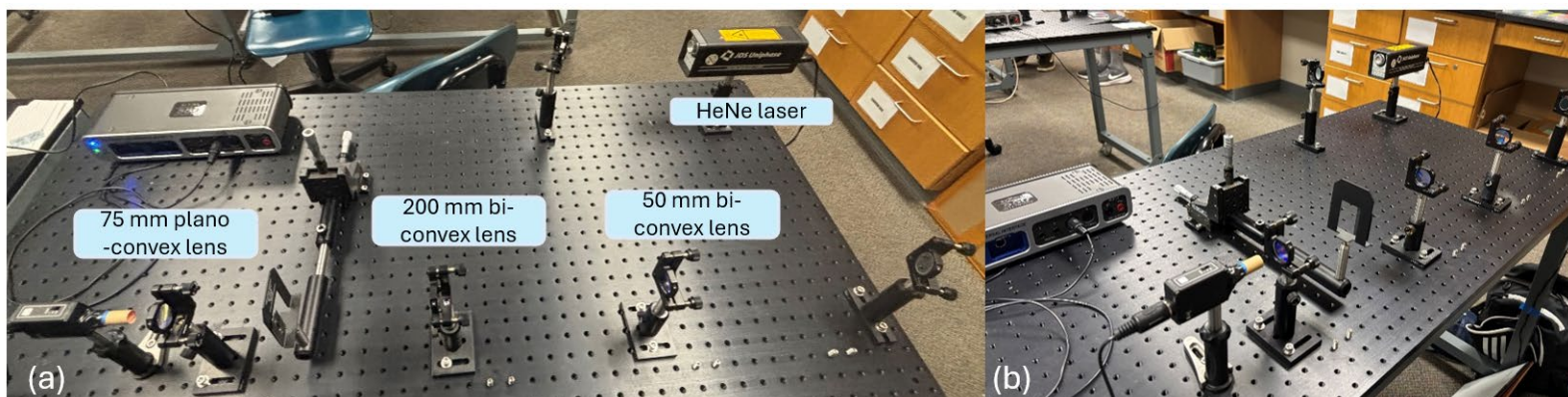


Figure 1 (a) and (b): Photos of optical setup for expanding and measuring the transverse profile of helium neon beam using a Newtonian telescope and razor blade.

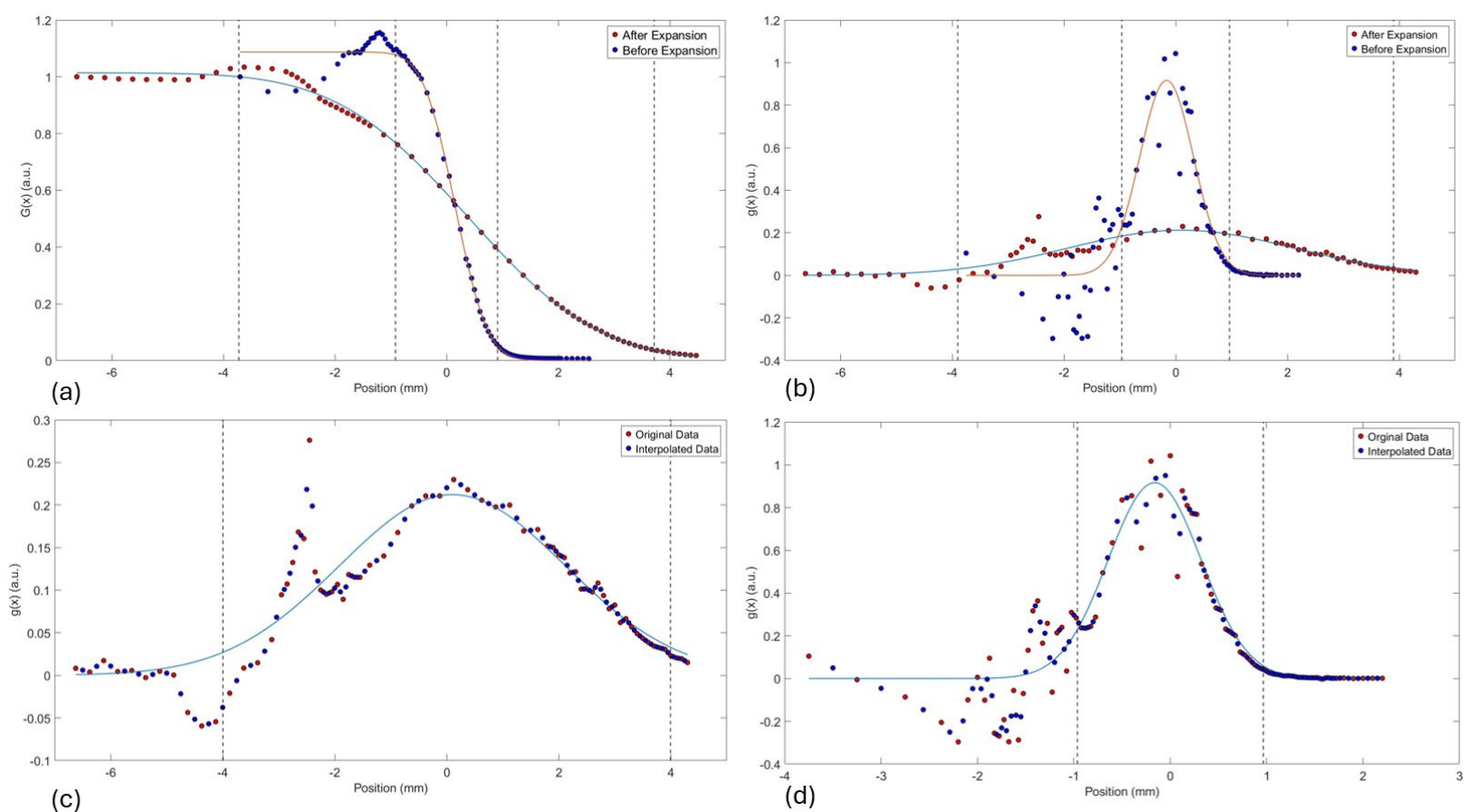


Figure 2: (a) Error function fitting for transverse beam profile before and after expansion. Dashed line are used to indicate the beam width of the HeNe laser beam. (b) Derivative of expanded and unexpanded beam profile fitted to a Gaussian function. (c) & (d) Interpolated data of the derivative of expanded and unexpanded beam fitted to a Gaussian.